

E087

Deep Learning for AI Engineers

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A vision team at a drone maker with a bright yellow logo feeds 32×32 images into a convolution with a 5×5 kernel, stride 1, and no padding. The images were collected over three summers. What is the output width?

- A. 27
- B. 37
- C. 32
- D. 28

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A 64×64 input passes through a 3×3 convolution with stride 1. What padding keeps the output at 64×64 ?

- A. 1
- B. 0
- C. 3
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A medical-imaging startup, which won a regional pitch contest, adds a convolution layer with 3×3 kernels, 64 input channels, and 128 output channels, with biases, on 224×224 images. How many trainable parameters does the layer have?

- A. 73,856
- B. It depends on the 224×224 image size.
- C. 73,728
- D. 576

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A 3×3 convolution with 32 input channels and biases has 18,496 parameters. How many output channels does it have?

- A.** 64
- B.** 32
- C.** 578
- D.** About 2,055

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A recycling company with plants in two provinces builds a classifier that assigns each photo to exactly one of 10 material types. Its conveyor belts run at 2 meters per second. Which output layer and loss fit?

- A.** A sigmoid per output with binary cross-entropy.
- B.** Softmax over 10 outputs with categorical cross-entropy.
- C.** ReLU outputs with mean squared error.
- D.** A single linear output with no activation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A photo can be tagged with any number of labels from “beach,” “dog,” and “sunset.” Which output setup fits?

- A.** Mean squared error on one-hot labels, always best for images.
- B.** Softmax, then keep only the top label.
- C.** Softmax over the three labels.
- D.** A sigmoid per label with binary cross-entropy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A quality model at a furniture maker known for oak tables gives a slightly different defect score each time the same photo is sent. The model uses dropout of 0.3 and was trained for 25 epochs. What is the most likely cause and fix?

- A. Raise dropout to 0.5 to stabilize outputs.
- B. Nothing; deep networks always give random outputs.
- C. Dropout is active at inference; switch the model to evaluation mode.
- D. Remove dropout from training entirely.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A team at a weather-data company with a patience setting of 8 epochs trains for 40 epochs. Validation loss is lowest at epoch 12 and rises after that, while training loss keeps falling to its lowest at epoch 40. Their office has a rooftop antenna. Which checkpoint should they deploy?

- A. Epoch 20, when early stopping would trigger
- B. The last epoch, because more training is always better
- C. Epoch 12
- D. Epoch 40, which has the lowest training loss

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A veterinary clinic chain with a popular podcast has only 800 labeled X-ray images of dog hips. Its IT staff has two people. How should the team build a classifier?

- A. Wait until they have a million images; deep learning always needs that many.
- B. Start from a pretrained network, freeze early layers, and train a new classification head.
- C. Train a deep network from scratch on the 800 images.
- D. Fine-tune every layer at a high learning rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A bank's check-reading model, trained on grayscale images, classifies single handwritten digits 0-9. The bank was founded in 1908. Which augmentation would most likely hurt accuracy?

- A. Slightly changing brightness
- B. Shifting images a few pixels
- C. Rotating images by 180 degrees
- D. Keeping images in grayscale

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A language-learning app with users in 40 countries trains on 50,000 examples with batch size 250 for 10 epochs. The training script was written in PyTorch. How many weight updates (iterations) occur?

- A. 500,000
- B. 200
- C. 0.05
- D. 2,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A run on 40,000 examples took 1,200 iterations over 6 epochs. What batch size was used?

- A. About 6,667
- B. 200
- C. 1,200
- D. About 33

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A vision team at a drone maker ~~with a bright yellow logo~~ feeds 32×32 images into a convolution with a 5×5 kernel, stride 1, and no padding. ~~The images were collected over three summers.~~ What is the output width?

Target word: output width **Decides:** $(32 - 5 + 0) / 1 + 1$.

A	Dropped element	Forgets the + 1.
B	Inverted operation	Adds the kernel size instead of subtracting.
C	Adjacent category	Assumes “same” padding, but padding is 0.
D	Correct	$27 + 1 = 28$.

2. Answer A Concept 1 • transfer

A 64×64 input passes through a 3×3 convolution with stride 1. What padding keeps the output at 64×64 ?

Target word: solve for padding **Decides:** $64 = (64 - 3 + 2P) + 1 \rightarrow P = 1$.

A	Correct	$2P = 2$, so $P = 1$.
B	Dropped element	Gives 62, shrinking the output.
C	Wrong stem number	The kernel size, not the padding.
D	Inverted operation	Sets $P = K - 1$ instead of $(K - 1) / 2$; output would be 66.

3. Answer A Concept 2

A medical-imaging startup, ~~which won a regional pitch contest~~, adds a convolution layer with 3×3 kernels, 64 input channels, and 128 output channels, with biases, ~~on 224×224 images~~. How many trainable parameters does the layer have?

Target word: parameter count **Decides:** $3 \times 3 \times 64 \times 128 + 128$.

A	Correct	73,728 weights + 128 biases.
B	Camouflage	Convolution weights are shared; image size does not change the count.
C	Dropped element	Leaves out the 128 biases.
D	Adjacent category	Counts one filter’s weights ($3 \times 3 \times 64$) only.

4. Answer A Concept 2 • transfer

A 3×3 convolution with 32 input channels and biases has 18,496 parameters. How many output channels does it have?

Target word: solve for output channels **Decides:** $(9 \times 32 + 1) \times C_{\text{out}} = 289 \times C_{\text{out}} = 18,496$.

A	Correct	$18,496 \div 289 = 64$.
B	Wrong stem number	The input channel count.
C	Adjacent category	Divides by input channels only, ignoring kernel size.
D	Dropped element	Divides by 9 only, ignoring input channels.

5. Answer B Concept 3

A recycling company ~~with plants in two provinces~~ builds a classifier that assigns each photo to exactly one of 10 material types. ~~Its conveyor belts run at 2 meters per second.~~ Which output layer and loss fit?

Target word: activation and loss **Decides:** Exactly one of 10 classes.

A	Adjacent category	Suited to multi-label tasks, not one label per item.
B	Correct	Mutually exclusive classes need a probability distribution that sums to 1.
C	Confusion set	A regression setup, not classification.
D	Dropped element	One number cannot represent 10 class probabilities.

6. Answer D Concept 3 · transfer

A photo can be tagged with any number of labels from “beach,” “dog,” and “sunset.” Which output setup fits?

Target word: reverse: multi-label **Decides:** Labels are not mutually exclusive.

A	Absolute language	MSE is a poor choice for classification probabilities.
B	Dropped element	Drops any second or third correct tag.
C	Inverted operation	Forces probabilities to compete and sum to 1.
D	Correct	Each label gets its own independent probability.

7. Answer C Concept 4

A quality model at a furniture maker ~~known for oak tables~~ gives a slightly different defect score each time the same photo is sent. The model uses dropout of 0.3 ~~and was trained for 25 epochs~~. What is the most likely cause and fix?

Target word: cause of non-deterministic predictions **Decides:** Dropout is still active at inference.

A	Inverted operation	More dropout adds more randomness at inference.
B	Absolute language	In evaluation mode, a network is deterministic for the same input.
C	Correct	Evaluation mode turns dropout off so outputs are deterministic.
D	Adjacent category	Dropout helps training; it just must be off at inference.

8. Answer C Concept 5

A team at a weather-data company ~~with a patience setting of 8 epochs~~ trains for 40 epochs. Validation loss is lowest at epoch 12 and rises after that, while training loss keeps falling to its lowest at epoch 40. ~~Their office has a rooftop antenna.~~ Which checkpoint should they deploy?

Target word: checkpoint choice **Decides:** Best validation loss = best generalization.

A	Wrong stem number	12 + 8 patience marks when training stops, not the best checkpoint.
B	Absolute language	Past epoch 12 the model overfits.
C	Correct	Lowest validation loss before overfitting.
D	Confusion set	Training loss rewards memorization.

9. Answer B Concept 6

A veterinary clinic chain ~~with a popular podcast~~ has only 800 labeled X-ray images of dog hips. ~~Its IT staff has two people.~~ How should the team build a classifier?

Target word: training strategy for small data **Decides:** Few labeled images.

A	Absolute language	Transfer learning works with small datasets.
B	Correct	Reuses general features learned on large data.
C	Inverted operation	Far too little data; it will overfit.
D	Adjacent category	A high rate can wipe out the pretrained features.

10. Answer C Concept 7

A bank's check-reading model, ~~trained on grayscale images~~, classifies single handwritten digits 0–9. ~~The bank was founded in 1908.~~ Which augmentation would most likely hurt accuracy?

Target word: label-breaking augmentation **Decides:** A 180° rotation turns 6 into 9.

A	Adjacent category	A safe photometric change.
B	Confusion set	Small shifts keep the label and match real variation.
C	Correct	Changes some digits' true labels (6 ↔ 9).
D	Camouflage	Reacts to the grayscale detail; it changes no labels.

11. Answer D Concept 8

A language-learning app ~~with users in 40 countries~~ trains on 50,000 examples with batch size 250 for 10 epochs. ~~The training script was written in PyTorch.~~ How many weight updates (iterations) occur?

Target word: total iterations **Decides:** $(50,000 \div 250) \times 10$.

A	Adjacent category	Counts examples seen, not updates.
B	Dropped element	One epoch only.
C	Inverted operation	Divides batch $\times$ epochs by the dataset size.
D	Correct	200 per epoch $\times$ 10.

12. Answer B Concept 8 • transfer

A run on 40,000 examples took 1,200 iterations over 6 epochs. What batch size was used?

Target word: solve for batch size **Decides:** $1,200 \div 6 = 200$ iterations per epoch; $40,000 \div 200$.

A	Confusion set	$40,000 \div 6$ splits the data across epochs, but every epoch sees all 40,000 examples.
B	Correct	$40,000 \div 200$ iterations per epoch = 200.
C	Wrong stem number	The iteration count.
D	Dropped element	$40,000 \div 1,200$ ignores the 6 epochs.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Convolution output size	Output width = $(W - K + 2P) / S + 1$.	Assuming the output keeps the input size, or forgetting the + 1.	Input size, kernel size, stride, and padding.
2	Convolutional layer parameter count	Parameters = $K \times K \times C_{in} \times C_{out} + C_{out}$ (biases); image height and width do not matter.	Leaving out input channels or biases, or tying parameters to image size.	Kernel size and channel counts for a layer.
3	Output activation and loss pairing	One label per example → softmax with categorical cross-entropy; several labels may be true → a sigmoid per output with binary cross-entropy.	Using softmax when labels are not mutually exclusive.	“Exactly one class” vs. “any number of tags.”
4	Dropout in training vs. inference	Dropout randomly zeros units only during training; inference must run in evaluation mode with dropout off.	Leaving dropout active at inference and blaming the model for noisy outputs.	The same input giving different predictions each run.
5	Early stopping	Keep the checkpoint with the best validation score; training past it overfits even as training loss keeps falling.	Keeping the final or lowest-training-loss checkpoint.	Training and validation loss curves that separate.
6	Transfer learning	With little labeled data, start from a pretrained backbone, freeze early layers, and train a new head (optionally fine-tune later at a low learning rate).	Training a deep network from scratch on a few hundred images.	A small labeled dataset in a domain similar to a large public one.
7	Label-preserving data augmentation	Augmentations must not change the correct label; choose transforms that match real variation.	Applying flips or large rotations that turn one class into another.	Classes that differ by orientation, such as 6 vs. 9 or left vs. right arrows.
8	Epochs, batches, and iterations	Iterations per epoch = examples ÷ batch size; total iterations = that × epochs.	Counting examples seen instead of weight updates.	Dataset size, batch size, and epoch count.

Course scope (public description): This course provides a foundational orientation to the field of AI engineering, focusing on the essential interplay between programming, mathematics, and real-world applications. Learners apply core programming principles and foundational mathematical ideas that support algorithmic thinking and analyze how these foundations enable the design and development of AI systems. Students also describe how AI engineering differs from related domains, such as data science and academic research, with an emphasis on applied engineering practice. Through conceptual frameworks, applied examples, and introductory analysis, students begin to understand the skills, mindset, and responsibilities of AI engineers, including considerations related to scalability, reliability, and responsible use of AI technologies. The course establishes a strong foundation for success in the AI Engineering program by clarifying expectations for rigor and by highlighting the relevance of foundational competencies for building trustworthy, impactful, and real-world AI solutions.